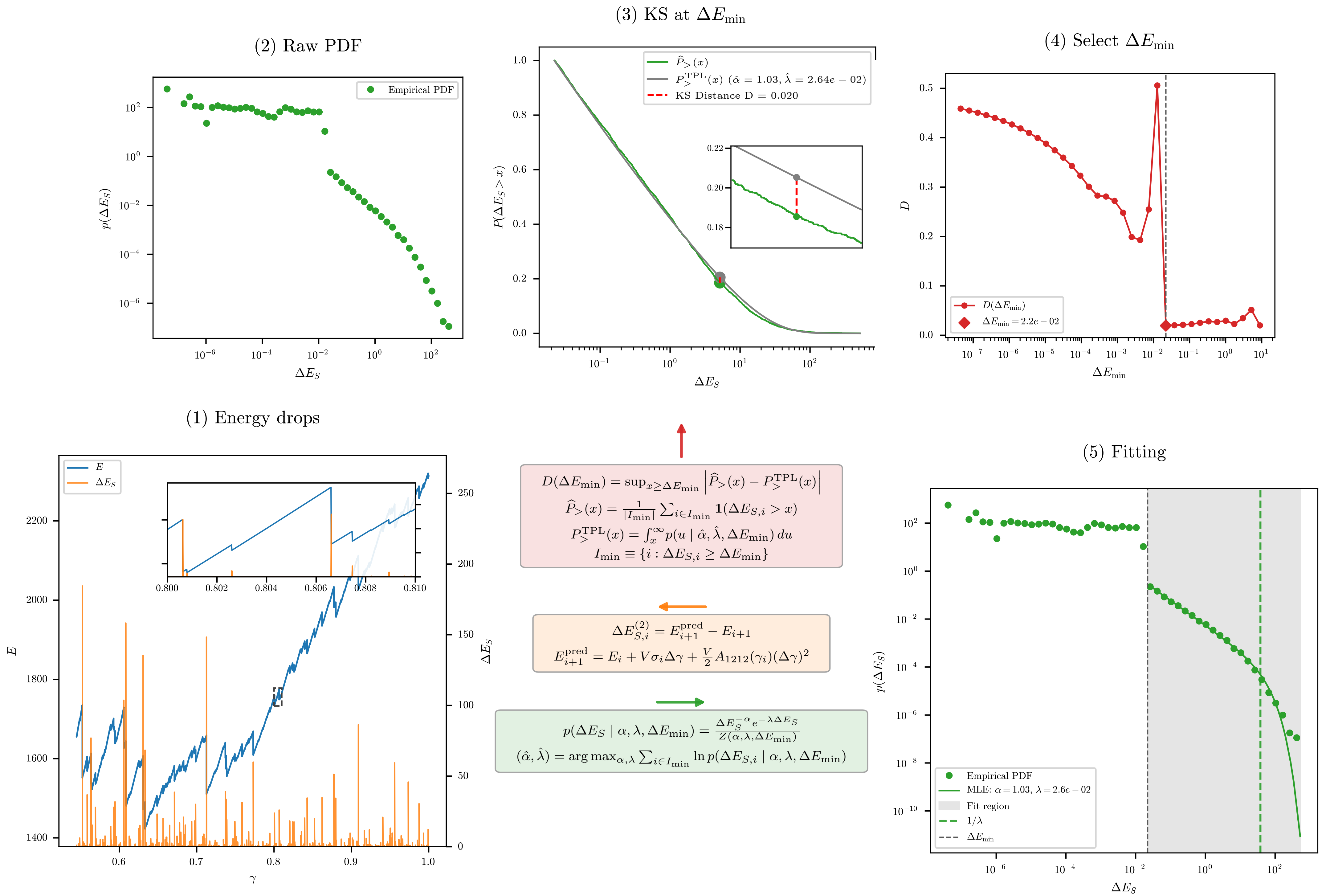
(1) Energy drops
$E$
$\Delta E_S$
$\gamma$
(2) Raw PDF
Empirical PDF
$p(\Delta E_S)$
$\Delta E_S$
(3) KS at $\Delta E_{\min}$
$\widehat{P}_>(x)$
$P_>^{\mathrm{TPL}}(x)$ $(\hat{\alpha} = 1.03, \hat{\lambda} = 2.64e-02)$
KS Distance D = 0.020
$P(\Delta E_S > x)$
$\Delta E_S$
(4) Select $\Delta E_{\min}$
$D(\Delta E_{\min})$
$\Delta E_{\min} = 2.2e-02$
$D$
$\Delta E_{\min}$
$D(\Delta E_{\min}) = \sup_{x \geq \Delta E_{\min}} \left| \widehat{P}_>(x) - P_>^{\mathrm{TPL}}(x) \right|$
$\widehat{P}_>(x) = \frac{1}{|I_{\min}|} \sum_{i \in I_{\min}} \mathbf{1}(\Delta E_{S,i} > x)$
$P_>^{\mathrm{TPL}}(x) = \int_x^\infty p(u \mid \hat{\alpha}, \hat{\lambda}, \Delta E_{\min})\, du$
$I_{\min} \equiv \{i : \Delta E_{S,i} \geq \Delta E_{\min}\}$
$\Delta E_{S,i}^{(2)} = E_{i+1}^{\mathrm{pred}} - E_{i+1}$
$E_{i+1}^{\mathrm{pred}} = E_i + V\sigma_i\Delta\gamma + \frac{V}{2} A_{1212}(\gamma_i)(\Delta\gamma)^2$
$p(\Delta E_S \mid \alpha, \lambda, \Delta E_{\min}) = \frac{\Delta E_S^{-\alpha} e^{-\lambda \Delta E_S}}{Z(\alpha, \lambda, \Delta E_{\min})}$
$(\hat{\alpha}, \hat{\lambda}) = \arg\max_{\alpha,\lambda} \sum_{i \in I_{\min}} \ln p(\Delta E_{S,i} \mid \alpha, \lambda, \Delta E_{\min})$
(5) Fitting
Empirical PDF
MLE: $\alpha = 1.03$, $\lambda = 2.6e-02$
Fit region
$1/\lambda$
$\Delta E_{\min}$
$p(\Delta E_S)$
$\Delta E_S$